Polynomial Multiplication via FFT

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1. Divide and Conquer.
2. DP
3. Greedy
4. NP-completeness

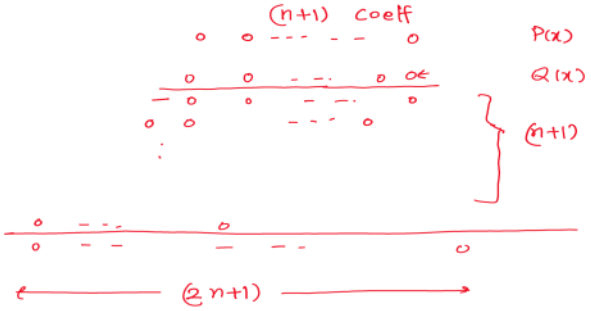
Polynomial Multiplication

$P(x) = 3x^2 + 8x + 5$

$Q(x) = 4x^2 + 9x + 8$

$P(x)\,Q(x) = 24x^4 + 64x \ldots 40$

Consider $P(x)$ and $Q(x)$ are of deg n



Each row take $\Omega(n)$ time, there are $(n+1)$ rows, overall $\Omega(n^2)$ ti

General Problem stmt

$A(x) = a_0 + a_1x + a_2x^2 + \ldots + a_{n-1}x^{n-1}$

$B(x) = b_0 + b_1x + b_2x^2 + \ldots + b_{n-1}x^{n-1}$

We aim to compute product $C(x) = A(x)B(x)$

$P(x) = [2x^2 + 5x + 9]$ $\quad P(x) = 27$ $\quad x = 2$ $\quad Q(x) = 11$

$Q(x) = [x^2 + x + 5]$ $\quad P(x) = 16$ $\quad x = 1$ $\quad Q(1) = 7$

$P(x) = $ $\quad x = 3$ $\quad Q(3) = 17$

$P(x) = [(2, 27)\ \ (1, 16)\ \ (3,)]$

Point value (evaluation of poly on the pt)

$Q(x) = [(2, 11)\ ,\ (1, 7)\ ,\ (3, 17)]$

$P(x)Q(x) = [(2, 11\times27)\ ,\ (1, 16\times7)\ ,\ (3,)]$

Point Value Representation Given pts $(x_0, x_1 \ldots x_{n-1}) \in \mathbb{C}^n$, then a poly $A(x)$ can be represented by the set

$\{(x_0, A(x_0)),\ (x_1, A(x_1)) \ldots (x_{n-1}, A(x_{n-1}))\}$ — representation of A

x_0 y_0

Using this representation, multiplication of two poly A, B become efficient

$\{(x_0, A(x_0)B(x_0)),\ (x_1, A(x_1)B(x_1)) \ldots (x_{n-1}, A(x_{n-1})\}$ — A(x)B(x)

But deg of $A(x)B(x)$ can be of $2n-2$.

But deg of $A(x)\,B(x)$ can be of $2n-2$.

Then need to evaluate $A(x)$ and $B(x)$ on $2n-2$ distinct pt and ei product

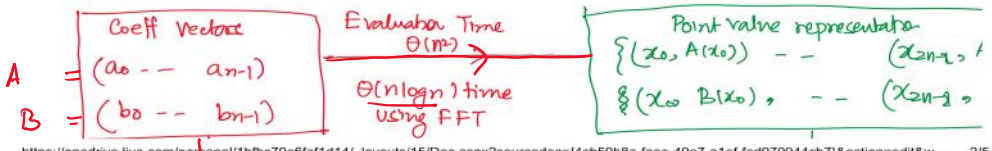
$A(x) = a_0 + a_1 x + a_2 x^2 - - + a_{n-1} x^{n-1}$ $\quad (a_0, a_1 - - a_{n-1})$

$x_0 - - x_{2n-2}$

$x_i = \omega_n$

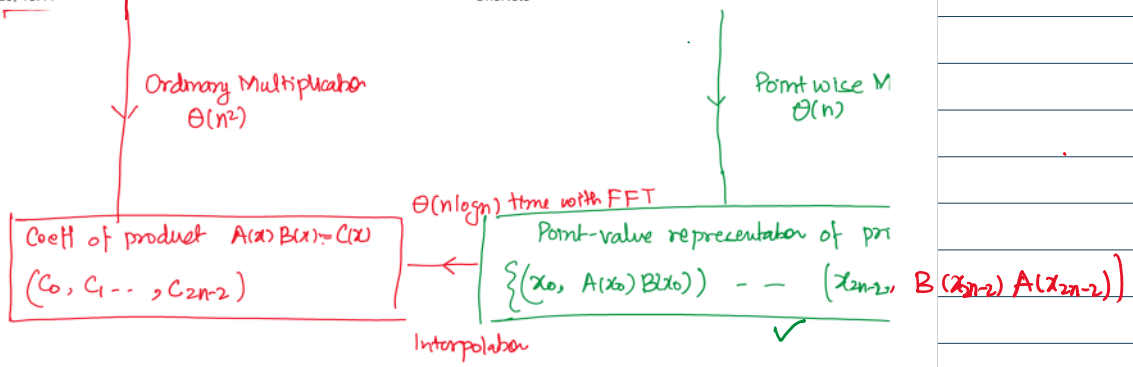
ω_{2n-1}^{k}

$k = 0 - - 2n-1$



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02/10/2026, 18:41 OneNote



Cube root of Unity $\quad x^3 = 1 \quad 1, \omega, \omega^2 \quad \omega = e^{i\pi/3}$

$1 + \omega + \omega^2 = 0, \quad \omega^3 = 1$

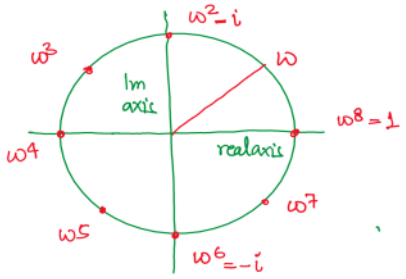
Complex n^{th} root of Unity

A number $z \in \mathbb{C}$ is n^{th} root of unity if $z^n = 1$

- principal n^{th} root of unity $\omega_n = e^{2\pi i/n}$ $\quad e^{i\theta} := \cos\theta$
- For $n \geq 1$, n^{th} root of unity are $1, \omega_n, \omega_n^2, \omega_n^3 - - \omega_n^{n-1}$

$\omega_n = e^{k2\pi i/n}$

$= \cos k\frac{2\pi}{n} + i \sin k\frac{2\pi}{n}$



$\omega_n :=$ principal n^{th} root of unity

$= e^{i2\pi/n}$

Lemma (Cancellation lemma) For integers $n > 0$, $k \geq 0$, $d > 0$

$$\omega_{dn}^{dk} = \omega_n^k$$

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Proof $\omega_{dn}^{dk} = \left(e^{\frac{i2\pi}{dn}}\right)^{dk} = e^{\frac{2\pi i}{dn}\cdot dk} = e^{\frac{2\pi i k}{n}} =$

Lemma (Summation lemma) For $n \geq 1$ and k is not divisible by n.

$$\sum_{j=0}^{n-1} (\omega_n^k)^j = 0$$

Proof $\sum_{j=0}^{n-1} (\omega_n^k)^j = \frac{(\omega_n^k)^n - 1}{\omega_n^k - 1} = \frac{(\omega_n^n)^k - 1}{\omega_n^k - 1} = \frac{1^k - 1}{\omega_n^k - 1}$

Since k is not divisible by n $\omega_n^k \neq 1$

Lemma (Halving lemma)

(a) $(\omega_n^k)^2 = \omega_{n/2}^k$

(b) $(\omega_n^{k+\frac{n}{2}})^2 = (\omega_n^k)^2$

Proof (a) $(\omega_n^k)^2 = \omega_n^{2k} = \omega_{n/2}^k$

(b) $(\omega_n^{k+\frac{n}{2}})^2 = \omega_n^{n+2k} = \omega_n^n \cdot \omega_n^{2k} = \omega_n^{2k} = (\omega_n^k)^2$

DFT $A(x) = \sum_{j=0}^{n-1} a_j x^j$ $a = (a_0, a_1, \ldots a_{n-1})$ coefficient

We evaluate $A(x)$ on $\omega_n^0, \omega_n, \omega_n^2, \ldots \omega_n^{n-1}$

$\hat{a}_k := A(\omega_n^k) = \sum_{j=0}^{n-1} a_j \omega_n^{kj}$ for $k = \{0 \ldots n-1\}$

The vector $\hat{a} := (\hat{a}_0, \hat{a}_1, \ldots \hat{a}_{n-1})$ is called as of the coeff vector $a = (a_0, a_1 \ldots a_{n-1})$

$A(x) = a_0 + a_1 x + a_2 x^2 \ldots + a_{n-1} x^{n-1}$ — (1)

$= (a_0 + a_2x^2 + a_4x^4 + \cdots + a_{n-2}x^{n-2}) + (a_1x + a_3x^3 + a_5$

$= (a_0 + a_2x^2 + a_4x^4 + \cdots a_{n-2}x^{n-2}) + x(a_1 + a_3x^2 + a_5$ — (2)

$A_e(x) = a_0 + a_2x + a_4x^2 + \cdots + a_{n-2}x^{\frac{n}{2}-1}$ — (2)

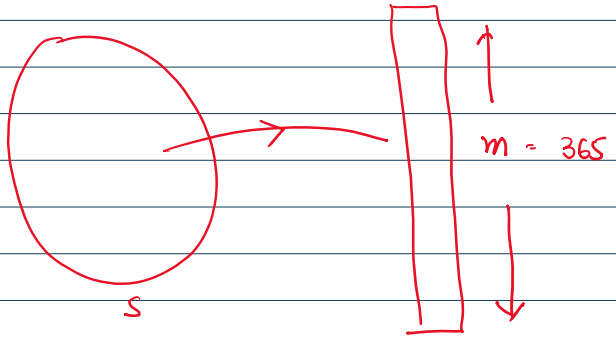
$A_o(x) = a_1 + a_3x + a_5x^2 + \cdots + a_{n-1}x^{\frac{n}{2}-1}$ — (3)

$A(x) = A_e(x^2) + x\,A_o(x^2)$ — (IV)

Due to (IV) the problem of evaluating A at $\omega^0, \cdots \omega^{n-1}$, following

- Evaluating two poly of deg $(\frac{n}{2}-1)$ at pts $(\omega^0)^2, (\omega^1)^2, \ldots$
- Combine the result using Eq IV

Q-2



$|S| = n$ given

A = Atleast two students have same birthday

P(A) = 1 − Pr (all birthday are different)

For first student $P_1 = \frac{365}{365} = 1$

2nd " $P_2 = \frac{364}{}$

$$\frac{365}{365}$$

2nd : $P_2 = \frac{364}{365}$

3rd : $P_3 = \frac{363}{365}$

nth : $P_n = \frac{365-(n-1)}{365}$

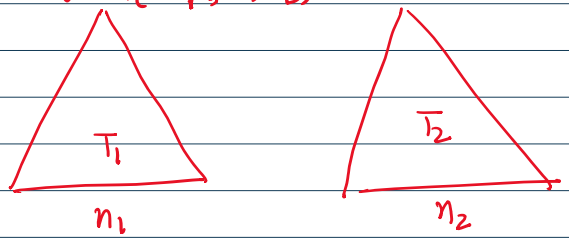
Pr all birthday are different $= \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365-(n-1)}{365}$

$$= \prod_{i=0}^{n-1} \frac{(365-i)}{365}$$

We want $P(A) \geq 1/2 \Rightarrow 1 - \prod_{i=0}^{n-1} \left(\frac{365-i}{365}\right) \geq 1/2$

$n \geq 23$

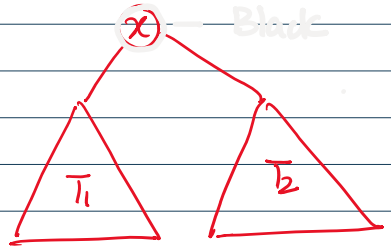
Que 6 RBJOIN(T_1, x, T_2)



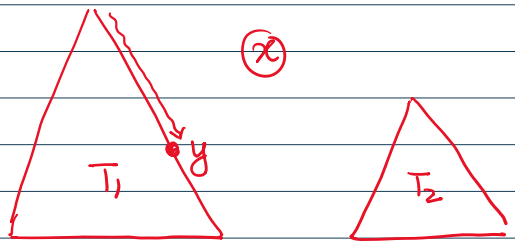
$k_1 < x < k_2$

$\forall k_1 \in T_1$ and $k_2 \in T_2$

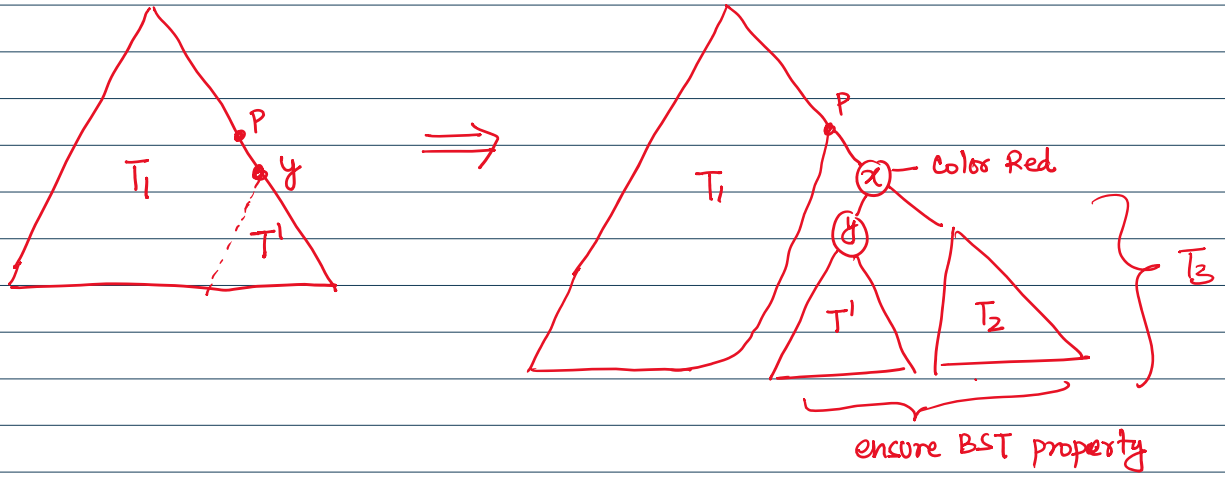
Case (1) $bh(T_1) = bh(T_2)$



Case (2) $bh(T_1) > bh(T_2)$

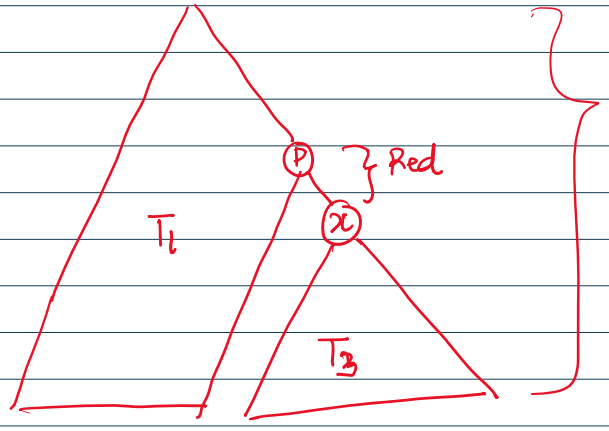


Find a node y s.t $bh(y) = bh(T_2)$



If color of p is black ⇒ no issue

If color of P is red ⇒ double red problem



Remaining step are same as RB- Insertion

Time complexity Finding y $O(\log n_1)$

Insertion in $T_1 \cup x \cup T_2$ $O(\log(n_1+n_2))$ overall $O(\log(n_1+n_2))$

Q2 Finding diameter

Step 1 Start from an arbitrary vertex $s \in V$

Run BFS and Find a vertex farthest from s

$$u = \underset{v \in V}{\operatorname{argmax}}\ \delta(s, v)$$

Fact 1 Vertex u is one of the end pt of diameter

Step Run BFS starting from vertex u

$v = \underset{w \in V}{\text{argmax}}\ \delta(u, w)$

Fact 2. v is other end pt of diameter.

Overall time complexity $O(|V|+|E|)$.

(End sem exclude only Asymptotic analysis + BST)

Algorithm

Recursive FFT (a) /* $a = (a_0, a_1 \dots a_{n-1})$ coeff of $A(x)$

$n == a.length$

if $(n == 1)$ then

return a

$a_e = (a_0, a_2 \dots a_{n-2})$ /* coeff corresponding A_e

$a_o = (a_1, a_3 \dots a_{n-1})$ " A_o

$y_e =$ Recursive-FFT (a_e)

$y_o =$ Recursive-FFT (a_o)

$\omega_n = e^{2\pi i/n}$ $\omega = 1$

For $k=0$ to $n/2 - 1$ do

$y_k = (y_e)_k + \omega (y_o)_k$

$y_{\frac{n}{2}+k} = (y_e)_k - \omega (y_o)_k$

$\omega \leftarrow \omega \cdot \omega_n$

$y = (y_0, y_1, \dots y_{n/2-1}, \dots y_n)$

return y

$(y_e)_k :=$ k^{th} index of y_e

$y_0 = A(\omega_n^0)$

$y_1 = A(\omega_n)$

$y_2 = A(\omega_n^2)$

Proof of correctness In for loop $k = 0 \dots n/2 - 1$

$(y_e)_k = A_e(\omega_{n/2}^k) = A_e(\omega_n^{2k})$

$(y_o)_k = A_o(\omega_{n/2}^k) = A_o(\omega_n^{2k})$

$y_0 = a_0 \cdot \omega^0 = a_0$

Proof of correctness In for loop $k = 0 \text{ -- } n/2 - 1$

$y_0 = a_0 \cdot \omega^0 = a_0$

$$(y_e)_k = A_e(\omega_{n/2}^k) = A_e(\omega_n^{2k})$$

$$(y_o)_k = A_o(\omega_{n/2}^k) = A_o(\omega_n^{2k})$$

For k from 0 to $n/2 - 1$

$$y_k = (y_e)_k + \omega^k (y_o)_k = A_e(\omega_n^{2k}) + \omega_n^k A_o(\omega_n^{2k}) = A(\omega_n^k) \quad \text{(3)}$$

$$y_{k+\frac{n}{2}} = (y_e)_k - \omega_n^k (y_o)_k$$

$$= A_e(\omega_n^{2k}) - \omega_n^k A_o(\omega_n^{2k})$$

$$= A_e(\omega_n^{2k+n}) + \omega_n^{k+\frac{n}{2}} A_o(\omega_n^{2k+n})$$

$$= A_e(\omega_{n/2}^{\frac{n}{2}+k}) + \omega_n^{\frac{n}{2}+k} A_o(\omega_{n/2}^{\frac{n+k}{2}})$$

$$= A(\omega^{k+n/2}) \quad \text{(IV)}$$

$-\omega_n^k = \omega_n^{n/2} \cdot \omega_n^k = \omega_n^{k+n/2}$

$\because \omega_n^{n/2} = -1$

$\omega^n = 1$

$\omega_n^{2k} = 1 \cdot \omega_n^{2k} = \omega_n^{n+2k}$

By (III) & (IV) we compute y_k $(0 \text{ --- } n-1)$

$T(n) = 2T(n/2) + \Theta(n)$ overall $T(n) = \Theta(n \log n)$

time required for evaluating $A(x)$ on $\omega_n^0 \text{ -- } \omega_n^{n-1}$

Interpolation via FFT

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & \omega_n & \cdots & \omega_n^{n-1} \\ \vdots & & & \\ 1 & \omega_n^{n-1} & \omega_n^{2(n-1)} \cdots & \omega_n^{(n-1)^2} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} y_0 \\ \vdots \\ y_{n-1} \end{bmatrix} = \begin{bmatrix} A(\omega_n^0) \\ A(\omega_n) \\ \vdots \end{bmatrix}$$

$= M_n(\omega_n)$

a = coeff of Ax

For interpolation step we have given $y = (y_0 \text{ -- } y_{n-1})$ $\{(x_0, A(x_0)B(x_0))$ ... $(x_{2n-2}, A(x_{2n-2})B(x_{2n-2})\}$

Need to compute $a = (a_0 \text{ -- } a_{n-1})$

$$\Rightarrow \begin{bmatrix} a_0 \\ \vdots \\ a_{n-1} \end{bmatrix} = \left[M_n(\omega_n)\right]^{-1} \begin{bmatrix} y_0 \\ \vdots \\ y_{n-1} \end{bmatrix}$$

$\omega = 1/\omega$

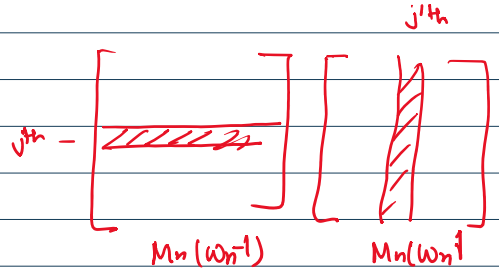
ω_{2n-1}

Theorem For $n \geq 1$, $M_n(\omega_n)$ is invertible

$$\left[M_n(\omega_n)\right]^{-1} = \frac{1}{n} M_n\left(1/\omega_n\right)$$

Proof (j,j') index of $M_n(\omega_n) = \omega_n^{jj'}$ $\quad -jj'$

" " $\frac{1}{n} M_n\left(1/\omega_n\right) = \frac{1}{n}\omega_n$



(j,j') index of $\frac{1}{n} M_n(\omega_n^{-1}) M_n(\omega_n) = \frac{1}{n} \sum_{k=0}^{n-1} \omega^{-kj} \cdot \omega^{kj'}$

$$= \frac{1}{n} \sum_{k=0}^{n-1} \omega^{k(j'-j)}$$

If $j \neq j'$ $\quad \frac{1}{n} \sum_{k=0}^{n-1} \omega_n^{k(j'-j)} = 0$ $\quad$ Non-diag are zero

If $j = j'$ $\quad \frac{1}{n} \sum_{k=0}^{n-1} \omega_n^{k \cdot 0} = 1$

Convolution $\quad x \in \mathbb{R}^n \quad y \in \mathbb{R}^n \quad x * y := z \in \mathbb{R}^{2n-1}$

$z_k = \sum_{j=0} x_j y_{k-j}$ $\quad x_j = j^{th}$ index of x

$z_0 = y_0 x_0$

$z_1 = x_0 y_1 + x_1 y_0$

$z_2 = x_0 y_2 + x_1 y_1 + x_2 y_0$

Exercise compute convolution in $O(n(\log n))$ time

Kronecker Product $x = (x_1, x_2, x_3)$ $y = (y_1, y_2, y_3)$

$x \otimes y = (x_1y_1, x_1y_2, x_1y_3 - - \quad x_2y_3)$

Ex. compute kronecker product in $\Theta(n \log n)$ time